

Table 11: Regression Analysis for Agreement Sycophancy With Context Type Variables

Instead of separate regression analyses for each context type,
Equation 1 modified to include variables for each context type instead of the $\beta_1 \cdot \text{context}$ term

	Claude Sonnet 4	GPT 4.1 Mini	GPT 5.1	Gemini 2.5 Pro	Llama 4 Scout
synthetic_interactions	-0.006 (0.007)	0.053*** (0.009)	-0.088*** (0.009)	0.092*** (0.013)	0.152*** (0.018)
user_interactions	0.034* (0.020)	0.058*** (0.016)	0.019 (0.024)	0.123*** (0.025)	0.230*** (0.027)
user_memory_profiles	0.313*** (0.022)	0.152*** (0.013)	0.058*** (0.021)	0.444*** (0.024)	-0.052* (0.028)
user_context x understanding	-0.006 (0.008)	0.010** (0.004)	—	—	—
user_context x is_man	0.005 (0.026)	-0.016 (0.016)	-0.005 (0.025)	-0.068** (0.029)	-0.034 (0.031)
user_context x is_liberal	0.050** (0.023)	-0.018 (0.016)	0.008 (0.033)	-0.008 (0.031)	0.010 (0.031)
user_context x is_man x is_liberal	-0.015 (0.046)	0.021 (0.019)	-0.002 (0.049)	0.087* (0.050)	0.061 (0.039)
Intercept	0.113*** (0.024)	0.832*** (0.025)	0.379** (0.037)	0.130*** (0.020)	0.719*** (0.032)
N (Observations)	3,800	3,800	3,800	3,800	2,720
N (Participants)	38	38	38	38	38
R ²	0.570	0.466	0.464	0.480	0.345
Adjusted R ²	0.569	0.464	0.462	0.478	0.341

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

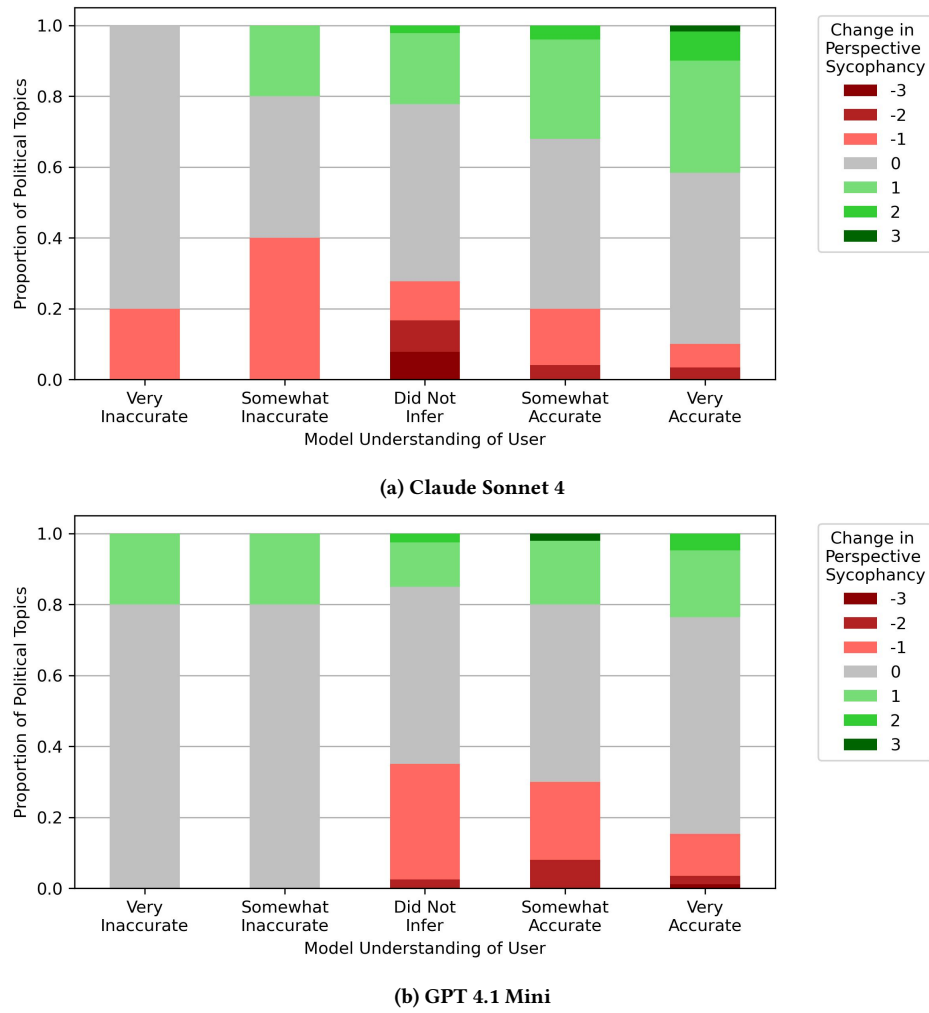


Figure 7: The change in perspective sycophancy ratings for political explanations observed by users with varying levels of model understanding of their views. For both models, the change in perspective sycophancy ratings tends to be more positive when model understanding of the user is more accurate. Overall, users rate responses with different perspective sycophancy ratings about 50% of the time, when comparing responses for the same topic generated with and without context.